Time limit: 15 minutes.
Instructions: This tiebreaker contains 3 short answer questions. All answers must be expressed in simplest form unless specified otherwise. You will submit answers to the problem as you solve them, and may solve problems in any order. You will not be informed whether your answer is correct until the end of the tiebreaker. You may submit multiple times for any of the problems, but **only the last submission for a given problem will be graded**. The participant who correctly answers the most problems wins the tiebreaker, with ties broken by the time of the last correct submission.
No calculators.

1. Quadrilateral $ABCD$ has side lengths $AB = 3$, $AD = 2$, and $CD = BC$. In addition, $\angle BAD = \angle ADC = 90°$. Compute the area of $ABCD$.

2. Consider a right cone of height and radius 1. Spin the cone around a line that passes through a point on the perimeter of the cone's base and is parallel to the cone's altitude. Compute the volume of the resulting solid.

3. In triangle ABC, two angle trisectors of A intersect BC at D and E such that D is closer to B and E is closer to C. Given that $BD = 3$, $DE = 1$ and $EC = 2$, compute $\angle DAE$ in degrees.